Week 1 Lesson 1: Introduction to Ethics, Professional Ethics and Ethical Frameworks

1. Introduction to Ethics

- **Definition:**
  Ethics is the branch of philosophy that deals with moral principles—what is right and wrong, good and bad.
- **Purpose:**
  Ethics guide how individuals behave and make decisions, especially in situations involving conflict, responsibility, or fairness.
- **Example:**
  Is it ethical to use someone's data without their consent?

2. What is Professional Ethics?

- **Definition:**
  Professional ethics refers to the moral standards and principles that guide behavior in the workplace and within a profession.
- **Why Important?**
  - **Maintains trust between professionals and clients/public**
  - **Promotes integrity, accountability, and competence**
  - **Ensures fairness and prevents abuse of power**
- **Examples:**
  - **A software engineer avoiding plagiarism or writing secure, bug-free code**
  - **A doctor maintaining patient confidentiality**

Ethics vs. Professional Ethics

Aspect	**Ethics**	**Professional Ethics**
Definition	**General moral principles about right and wrong**	**Ethical standards specific to a profession**
Applies To	**All aspects of life (personal, social, etc.)**	**Workplaces and professional behavior**

Source	**Culture, religion, philosophy, personal values**	**Codes of conduct, industry standards, professional bodies**
Examples	**Being honest with a friend**	**Keeping client information confidential in a law firm**
Purpose	**Guide personal and social behavior**	**Maintain trust, integrity, and accountability in a job**
Enforceability	**Usually self-regulated or socially expected**	**Often enforced by professional organizations or law**

Simple Definitions:

- **Ethics:**
  The general principles that help us decide what is right and wrong in everyday life.
- **Professional Ethics:**
  The set of rules or guidelines that govern how professionals should act in their specific field.

Examples:

- **Ethics: Not lying to a friend even when it's easier to do so.**
- **Professional Ethics: A software engineer refuses to copy another company's code because it violates intellectual property laws.**

3. Ethical Frameworks Overview

Ethical frameworks help professionals evaluate situations and make ethical decisions. Some common frameworks include:

- **Deontological Ethics (Duty-Based Ethics)**
- **Consequentialism (Utilitarianism)**
- **Virtue Ethics**
- **Rights-Based Ethics**

Ethical Frameworks Overview

Ethical frameworks help individuals make consistent, fair, and moral decisions—especially in professional settings. Let's explore the four most common types:

1. Deontological Ethics (Duty-Based Ethics)

- **What is it?**
  This framework focuses on doing your duty and following rules, no matter the outcome.
- **Key Idea:**
  An action is right if it follows a moral rule or duty, even if it has a bad result.
- **Example:**
  A doctor refuses to lie to a patient about their illness, even if the truth may cause distress—because honesty is a moral duty.

2. Consequentialism (Utilitarianism)

- **What is it?**
  This focuses on the outcome of actions. The best action is the one that brings the greatest good for the greatest number.
- **Key Idea:**
  The end justifies the means.
- **Example:**
  A company may choose to lay off 10 employees to save the jobs of 100 others—because the overall outcome is better for more people.

3. Virtue Ethics

- **What is it?**
  This framework emphasizes the character of the person making the decision, rather than rules or consequences.
- **Key Idea:**
  A good person will naturally do good things. Focus on moral character traits like honesty, courage, and kindness.
- **Example:**
  A teacher helps a struggling student out of compassion and a desire to be kind, not because of rules or results.

4. Rights-Based Ethics

What is it?

- **This framework focuses on the rights of individuals, such as the right to privacy, freedom, and safety.**
- **Rights-Based Ethics is a moral theory in which individual rights—such as privacy, freedom, security, etc.—hold central importance.**
- **According to this theory, an action is moral only when it respects and protects a person's fundamental rights.**

Key Ideas:

- **Every human being has certain inalienable and indivisible rights.**
- **Even if violating these rights produces good outcomes, the action is not moral.**
- **This ethical framework establishes moral principles or boundaries.**

Common Rights:

- **Right to Privacy**
- **Freedom of Expression**
- **Right to Life and Safety**
- **Right to Information**
- **Right to Equality before Law**

Key Idea:

- **An action is wrong if it violates someone's rights, even if it benefits others.**

Example:

- **A business cannot use customer data without consent—even if it would help improve their services—because it violates the customer's right to privacy.**

Framework	Focus	Key Principle	Example
Deontological	Duty / Rules	Follow moral rules no matter the outcome	Tell the truth even if it hurts
Consequentialism	Results / Outcomes	Do what brings the most good	Fire a few to save many jobs
Virtue Ethics	Character / Virtues	Be a good person with good intentions	Help others out of kindness
Rights-Based	Individual Rights	Respect others' rights	Don't use someone's data without permission

Utilitarian Ethics (Outcome-based Ethics)

Definition:

- **Utilitarianism focuses on the consequences or outcomes of actions. An action is considered ethical if it results in the greatest good for the greatest number.**
- **Utilitarianism is a moral theory in which the morality of an action is judged based on its results or consequences.**

Key Ideas:

- **"Greatest good for the greatest number" — an action is moral if it benefits the largest number of people.**
- **The outcome of an action is more important than the intention behind it.**

Principles:

- **If an action promotes the overall well-being of more people, it is moral.**
- **Even if some individuals are harmed, the action may still be acceptable if it benefits the majority.**
- **This ethical approach is often policy-neutral and context-dependent.**

Example:

- **Suppose a software company develops a system that saves time and money for one million users, but a few employees may lose their jobs as a result.**
- **From the utilitarian perspective, since the majority benefits, this could be considered a moral decision.**

Key Principle:

- **Maximize benefit, minimize harm.**

Examples in Software Engineering:

- **Example 1: A software company releases a critical patch quickly—even though it hasn't been fully tested—to prevent a widespread cyberattack. *Outcome: Protects millions of users from potential threats.***
- **Example 2: A developer collects anonymized user data to improve app functionality. Although there's some privacy trade-off, the app becomes significantly more helpful for users.**
- **Example 3: Introducing AI to automate tasks in a software company leads to layoffs, but significantly improves national productivity and reduces long-term costs for customers.**

- **Data Breach Disclosure: A company promptly discloses a breach to prevent harm to users, even if it damages its reputation.**
- **AI Bias Mitigation: Developers redesign an AI algorithm that unfairly rejects loan applications—improving fairness for all users.**
- **Bug Patch: Releasing a quick hotfix with minor risks to fix a major flaw affecting millions.**

Aristotelian Ethics (Virtue Ethics)

Definition:

- **Aristotelian ethics emphasize the character and moral virtues of the individual. Actions are ethical if they are in line with virtues like honesty, courage, integrity, fairness, and wisdom.**
- **According to Aristotle, an action is moral or right when it aligns with the virtues of a person who has a good and ethical character.**
- **He stated that humans should follow the "golden mean" in life — that is, developing balanced behavior by avoiding both excess and deficiency.**

Virtues:

- **Honesty**
- **Courage**
- **Wisdom**
- **Justice**
- **Self-control**
- **Integrity**

Key Principle:

- **Be a good person, not just do good things.**

Examples in Software Engineering:

- **Example 1: A developer refuses to manipulate user reviews for a product, even though it would benefit the company—because it's dishonest.**
- **Example 2: A software engineer admits to a security flaw they introduced, taking responsibility and working overtime to fix it.**
- **Example 3: A tech team chooses to make their software accessible to users with disabilities—not for profit, but because it's the right and fair thing to do.**

Refusing Unethical Orders: A developer refuses to insert hidden spyware into a product.

Transparency: An engineer openly documents limitations in an AI model, even though it might hurt adoption.

Inclusive Design: A team goes the extra mile to design for users with disabilities—not due to legal obligation, but ethical conviction.

Comparison Table:

Aspect	**Utilitarian Ethics**	**Aristotelian Ethics**
Focus	**Outcomes/Consequences**	**Moral character/virtues**
Key Question	**"What action gives best result?"**	**"What would a virtuous person do?"**

Decision Driver	**Utility**	**Integrity & virtue**

When Do We Use Which?

Situation	Utilitarian Approach	Virtue-Based Approach
Security patching	**"Fix fast for user safety"**	**"Be honest about risks introduced"**
Data privacy	**"Balance benefit vs. harm"**	**"Respect user dignity and trust"**
AI design	**"Maximize fairness and accuracy"**	**"Avoid harm even when it's legal"**
Workplace decisions	**"Best decision for the team"**	**"Act with empathy and fairness"**

Summary: Why Both Matter

- **Utilitarian Ethics helps us make efficient and scalable decisions in high-impact systems.**
- **Aristotelian Ethics builds moral responsibility and personal integrity within the profession.**
- **Together, they create a balanced ethical foundation for software engineers in a complex, fast-moving world.**